

LESSON 15

INTRODUCTION TO GRAPHS

Overview:

This lesson is designed to develop students' modeling skills by introducing them to graphical representations of real-world data, maps and graphs.

Learning Outcomes:

At the end of this lesson, the students can:

1. describe a graph;
2. distinguish a Euler circuit from Euler path;
3. solve problems on Euler circuits and paths;
4. find a Hamiltonian circuit of path in weighted graphs; and
5. apply Euler circuit and path and Hamiltonian circuit in real-life scenarios.

Materials Needed:

- Module (Printed/Softcopy)
- Internet Connection
- Smartphone/Laptop, Google Classroom, FB Group Chat and YouTube videos
- Calculator
- Graphing Notebook/Paper

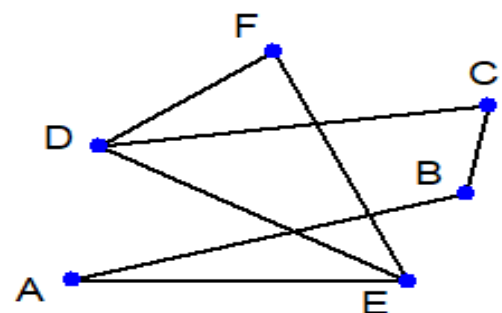
Duration: 6 hours

Learning Contents:

15.1 GRAPH

A **graph** is a set of point called **vertices** and line segments curves called **edges** that connect vertices. If two or more edges connect the same vertices, they are called **multiple edges**. If an edge begins and ends at the same vertex, it is called a **loop**.

Figure 16.1 is an example of a graph with 6 vertices. This graph may represent the flow of



the game with 6 teams (the vertices) and an edge connecting two vertices might mean that the two played against each other.

EXAMPLE 15.1: The Mathematics department is having difficulties courses A-G because of limited room availability. In the chart below, an “X” means that two courses cannot be scheduled at the same time.

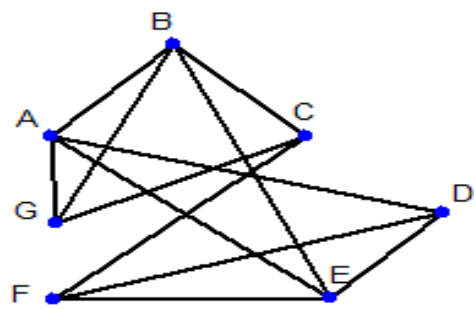
Figure 16.1

	A	B	C	D	E	F	G
A		X		X	X		X
B	X		X		X		X
C		X				X	X
D	X				X	X	
E	X	X		X		X	
F			X	X	X		
G	X	X	X				

Make a graph with vertices A-G. Make an edge between vertices if the line corresponding courses cannot be scheduled at the same time.

Solution:

The math department is having difficulties courses A-G because of limited room availability. The graph below represents the scheduling conflicts.



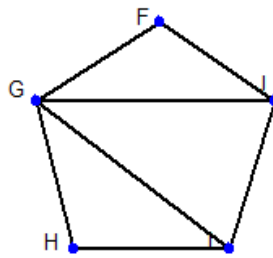
15.2 CONNECTED GRAPH AND COMPLETE GRAPH

A graph is either **connected** or **disconnected**. A graph is **connected** if any vertex can be reach from any other vertex by tracing along the edges. A connected graph in which every possible edge is drawn between vertices (without any multiple edges) is called a **complete graph**.

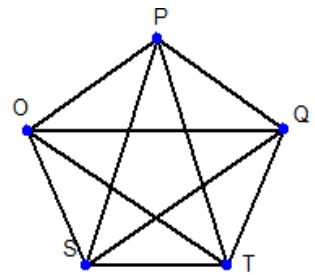
Here are some examples of graph.



This graph has five vertices but no edges. Hence this graph is disconnected.



This is a connected graph having five vertices with 7 edges and each vertex can be reached by tracing along.



This is an example of a complete graph with 5 vertices with 10 edges and each vertex can be reached by tracing along.

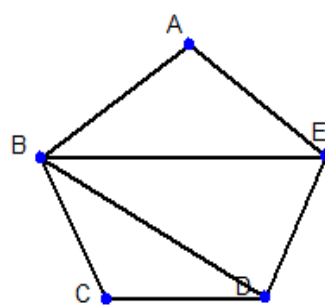
15.3 EULER CIRCUITS AND EULER PATHS

Euler Circuits

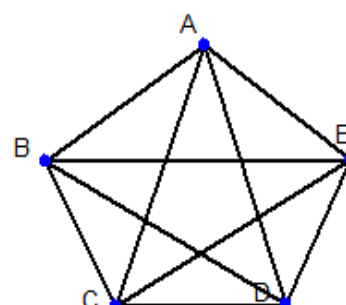
A **Euler circuit** is a circuit that uses every edge of the graph at exactly once. It starts and ends at the **same** vertex. Moreover, a graph will contain a Euler circuit if all vertices have even degree.

A **Euler path** is a path that uses every edge of the graph at exactly once. It starts and ends at **different** vertices. Furthermore, a graph will contain a Euler path if it contains utmost two vertices of odd degree.

EXAMPLE 15.2: Which of the following graph has a Euler circuit?



(a)

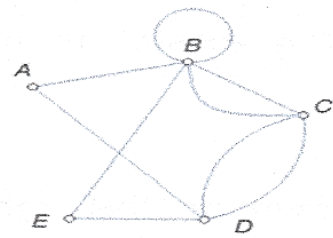


(b)

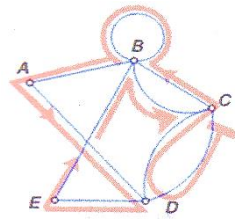
Solution:

- Vertices D and E are of odd degree. Thus, the does not have a Euler circuit. However, the graph has a Euler path.
- Each of the vertices is of even degree. Hence, the graph has a Euler circuit.

EXAMPLE 15.3: Determine whether the graph shown at the right has a Euler circuit or not.

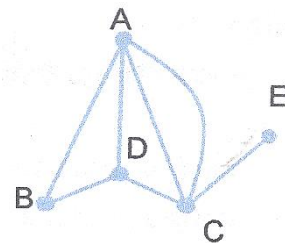


Solution:

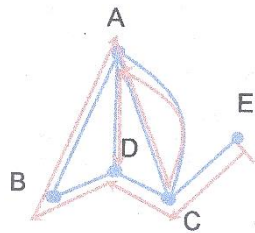


The graph has a Euler circuit CDCBBADEBC.

EXAMPLE 15.4: Does the graph (at the right) have a Euler circuit?

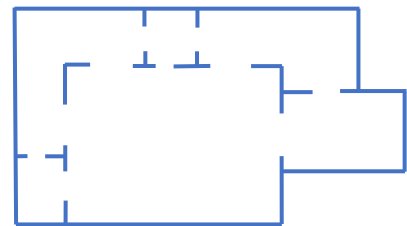


Solution:



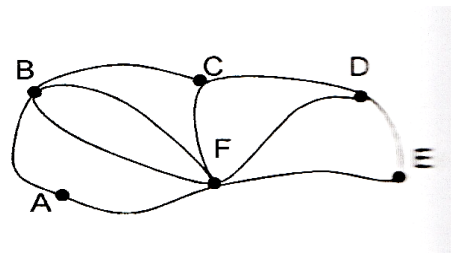
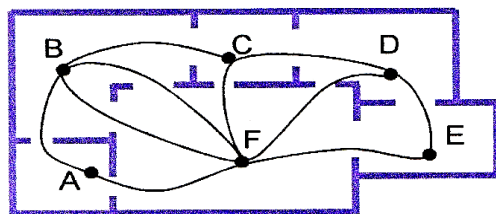
The graph does not have a Euler circuit but a Euler paths. One of these Euler paths is ECDBACAD.

EXAMPLE 15.5: The figure at the right depicts the floor plan of John’s house. Draw a graph that represents the floor plan, where vertices correspond to the rooms and edges correspond to doorways. Can John walk through every doorway exactly once to make a Euler circuit? Can he walk through every doorway exactly once to make a Euler path?



Solution:

As seen on the floor plan, the house has five rooms. Hence, it also has 5 vertices as follows.



There are two vertices of odd degree. Thus, we know that a Euler path exists but no Euler circuit. By trial and error, we can have a Euler path **CBFB AFEDCFD**.

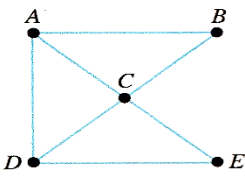
15.4 HAMILTONIAN CIRCUITS

A graph is **Hamiltonian** if there is a circuit that passes through each vertex at exactly once. This path is called **Hamiltonian circuit**. Just like Eulerian graphs, the Hamiltonian circuit (if it exists) usually NOT unique. Below are examples of Hamiltonian circuit.

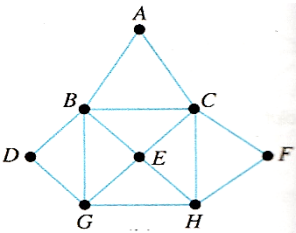
EXAMPLE 15.6: The graph shown on the right has Hamiltonian circuit. ABCEDA. Can you find them?

Solution:

Another Hamiltonian circuit is **BCEDAB**.



EXAMPLE 15.7: Find a Hamiltonian of the graph at the right.



Solution:

One of the Hamiltonian graph is **ABDGEHFCA**.

15.5 WEIGHTED GRAPHS

A **Weighted** graph is a graph in which each edge is associated with a value, called **weight** as shown in Figure 16.1. Values in a weighted graph can represent any quantity we desire. Note that the length of the edge does not necessarily correlate to its weight.

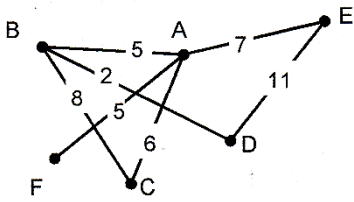
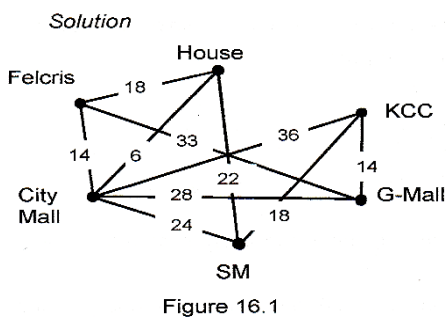


Figure 16.1

EXAMPLE 15.8: A shopper would like to go shopping at six different shopping malls tomorrow and then return home. The number of minutes spent traveling by vehicle between malls is given in the table. (N/A in the table indicates that no convenient vehicle route is available.) find one route for the tourist to follow and compute the travel time he need for that particular route.

LOCATION	House	Felcris	City Mall	SM	G-Mall	KCC
House	-	18	6	22	N/A	N/A
Felcris	18	-	14	N/A	33	N/A
City Mall	6	14	-	24	28	36
SM	22	N/A	24	-	N/A	18
G-Mall	N/A	33	28	N/A	-	14
KCC	N/A	N/A	36	18	14	-

Solution:



A route that visits each shopping mall just once corresponds to the Hamiltonian circuit. Beginning at the house, one such circuit is **house SM KCC G-Mall City Mall Felcris house**. By adding the weights of each edge in the circuit, we see that the total number of minutes the tourist will be spending is minutes.

Learning Activity:

- I. For each of these vertex-edge graphs, try to trace it (without lifting your pen from the paper, and without tracing any edge twice). If you succeed, number the edge in the order you used them (putting on arrows is optional), and circle whether you found a Euler circuit or a Euler path. The first one is done for you. Discuss your answer.

a.

a. Euler circuit
b. Euler path
c. Neither

b.

a. Euler circuit
b. Euler path
c. Neither

c.

a. Euler circuit
b. Euler path
c. Neither

d.

a. Euler circuit
b. Euler path
c. Neither

e.

a. Euler circuit
b. Euler path
c. Neither

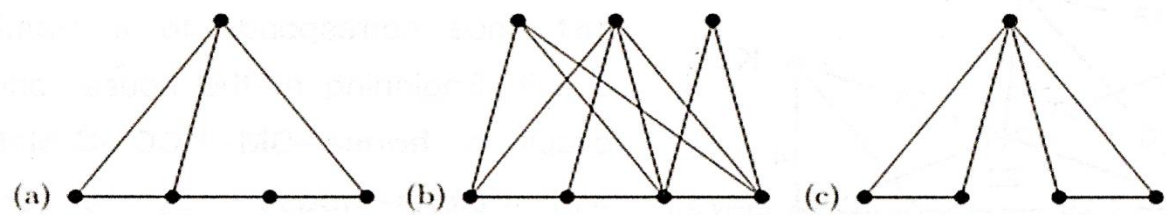
f.

a. Euler circuit
b. Euler path
c. Neither

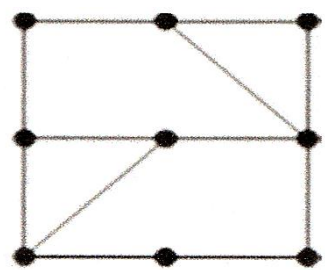
- II. Which of the six graphs above contains a Hamiltonian circuit? Explain your answer.

Learning Evaluation:

- 1) For each of the following graphs, decide whether or not they have a Euler circuit. For graphs with a Euler circuit, number the edges in the order of their appearance in a Euler circuit. If there is no Euler circuit, explain why not and state whether or not the graph has a Euler path.



- 2) Can you find a Hamiltonian circuit among the three graphs above? If yes, show how you find it.
- 3) Add edges until the following graph contains a Euler circuit. How many edges must you add? Explain your answer.



References:

Banigon Jr., R. et. Al. (2016). *Statistics and Probability*. Educational Resources Corporation. Cubao Quezon City.

Banigon Jr., R. et. Al. (2016). *General Mathematics*. Educational Resources Corporation. Cubao Quezon City.

Aufman, R. et. Al. (2018). *Mathematics in the Modern World*. Rex Book Store Inc. Sampaloc, Manila, Philippines.